

Implementation of removable singularities of $I_{\mathcal{N}}$

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The numerical piece of the calculation of the finite volume triangle function $\mathcal{G}$ involves the numerical evaluation of a 3-dimensional integral, which integrand was designed to be a smooth and convergent function. The details given in Ref. [1]. However, this is achieved by the careful cancellation of singularities in different terms. Here we derive the limit of the integrand whenever individual terms have singularities, and a direct numerical evaluation would be problematic. This would allow the code to directly evaluate the integrand near the singular points.

I. DEFINITIONS

The integral that we want to analyze was derived and described in Ref. [1],

$$\begin{aligned} \mathcal{I}_{\mathcal{N};\nu_1\cdots\nu_N}(\bar{\alpha}, P_f, P_i) \equiv & \int \frac{d^3\mathbf{k}}{(2\pi)^3} [H(\bar{\alpha}, \mathbf{k}) - 1] \sum_{j=0}^{n_j} c_j \left[D(\Lambda_j, \mathbf{k}) K_{\nu_1\cdots\nu_N}^\omega(\Lambda_j, \mathbf{k}) + \mathcal{K}_{r;\nu_1\cdots\nu_N}(\Lambda_j, \mathbf{k}) \right] \\ & - \int \frac{d^3\mathbf{k}}{(2\pi)^3} H(\bar{\alpha}, \mathbf{k}) \sum_{j=1}^{n_j} c_j D(\Lambda_j, \mathbf{k}) K_{\nu_1\cdots\nu_N}^\omega(\Lambda_j, \mathbf{k}) \\ & - \int \frac{d^3\mathbf{k}}{(2\pi)^3} H(\bar{\alpha}, \mathbf{k}) \sum_{j=0}^{n_j} c_j \mathcal{K}_{r;\nu_1\cdots\nu_N}(\Lambda_j, \mathbf{k}). \end{aligned} \quad (1)$$

$$H(\bar{\alpha}, \mathbf{k}) \equiv e^{-\bar{\alpha}(k_i^{*2} - q_i^{*2})(k_f^{*2} - q_f^{*2})}, \quad (2)$$

$$D(m, \mathbf{k}) \equiv \frac{1}{2\omega_{k2}} \frac{1}{(P_f - k)^2 - m_1^2 + i\epsilon} \frac{1}{(P_i - k)^2 - m_1^2 + i\epsilon} \Big|_{k^0 = \omega_{k2}}, \quad (3)$$

$$\omega_{k2} \equiv \sqrt{\mathbf{k}^2 + m_2^2}, \quad \omega_{Pk1} \equiv \sqrt{(\mathbf{P} - \mathbf{k})^2 + m_1^2}, \quad (4)$$

$$K_{\mu_1\cdots\mu_n}^\omega(m, \mathbf{k}) \equiv k_{\mu_1} \cdots k_{\mu_n} \Big|_{k^0 = \omega_{k2}}, \quad (5)$$

$$\mathcal{K}_{r;\nu_1\cdots\nu_N}(\Lambda, \mathbf{k}) \equiv D_{rf}(\Lambda, \mathbf{k}) \mathcal{N}_{\nu_1\cdots\nu_N}(E_f + \omega_{Pfk1}, \Lambda, \mathbf{k}) + D_{ri}(\Lambda, \mathbf{k}) \mathcal{N}_{\nu_1\cdots\nu_N}(E_i + \omega_{Pik1}, \Lambda, \mathbf{k}), \quad (6)$$

$$D_{rf}(m, \mathbf{k}) = \frac{1}{2\omega_{Pfk1}} \frac{1}{(E_f + \omega_{Pfk1})^2 - \omega_{k2}^2} \frac{1}{(E_i - E_f - \omega_{Pfk1})^2 - \omega_{Pik1}^2}, \quad (7)$$

$$D_{ri}(m, \mathbf{k}) = \frac{1}{2\omega_{Pik1}} \frac{1}{(E_i + \omega_{Pik1})^2 - \omega_{k2}^2} \frac{1}{(E_f - E_i - \omega_{Pik1})^2 - \omega_{Pfk1}^2}. \quad (8)$$

$$\mathcal{N}_{\nu_1\cdots\nu_N}(\Omega_k, \Lambda, \mathbf{k}) \equiv k_{\nu_1} \cdots k_{\nu_N} \Big|_{k^0 = \Omega_k} \quad (9)$$

Here we specialize in the following case

- Number of indices is $N \leq 1$, which will be useful for a transition of an S -wave system via a vector current.
- For these number of indices no subtractions are needed, i.e. $n_j = 0$.
- As described in appendix B6 of Ref. [1], this function is Lorentz covariant, so that we choose to evaluate it in the initial center of mass frame, with the final momentum aligned with the z -axis, such that

$$[\Lambda_{\beta_i}]^\mu{}_\nu P_i^\nu = (E_i^*, \mathbf{0}), \quad [\Lambda_{\beta_i}]^\mu{}_\nu P_f^\nu = (E_{f,i}^*, \mathbf{P}_{f,i}^*), \quad [R_{\hat{\mathbf{P}}_{f,i}}^{-1} \Lambda_{\beta_i}]^\mu{}_\nu P_f^\nu = (E_{f,i}^*, |\mathbf{P}_{f,i}^*| \hat{\mathbf{z}}), \quad (10)$$

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where we can describe the kinematics in terms of the Lorentz invariants $s_i = P_i^2$, $s_f = P_f^2$ and $\gamma = P_i \cdot P_f / \sqrt{s_i s_f}$, or alternatively the squares of P_i , P_f , and $q = P_f - P_i$.

- This means that none of the propagators will carry any azimuthal dependence, and the azimuthal integral can be carried out analytically, only taking care of the numerator terms.

The scalar integral simplifies to

$$\mathcal{I}_{\mathcal{N}}(\bar{\alpha}, P_f, P_i) \equiv \int \frac{d(\cos \theta_i^*) d|\mathbf{k}_i^*| \mathbf{k}_i^{*2}}{(2\pi)^2} [H(\bar{\alpha}, \mathbf{k}_i^*) - 1] \left[D(m, \mathbf{k}_i^*) + D_{rf}(m, \mathbf{k}_i^*) + D_{ri}(m, \mathbf{k}_i^*) \right] \\ - \int \frac{d(\cos \theta_i^*) d|\mathbf{k}_i^*| \mathbf{k}_i^{*2}}{(2\pi)^2} H(\bar{\alpha}, \mathbf{k}_i^*) (D_{rf}(m, \mathbf{k}_i^*) + D_{ri}(m, \mathbf{k}_i^*)). \quad (11)$$

The vector integral will be given by

$$\mathcal{I}_{\mathcal{N}}^\mu(\bar{\alpha}, P_f, P_i) = [(R_{\hat{\mathbf{p}}_{f,i}} \Lambda_{\beta_i})^{-1}]^\mu{}_\nu \mathcal{I}_{\mathcal{N}}^\nu(\bar{\alpha}, s_f, s_i, \gamma) \quad (12)$$

where the quantity on the right hand side is zero except for $\mu = 0$ and $\mu = 3$,

$$\mathcal{I}_{\mathcal{N}}^\mu(\bar{\alpha}, s_f, s_i, \gamma) \equiv \int \frac{d(\cos \theta_i^*) d|\mathbf{k}_i^*| \mathbf{k}_i^{*2}}{(2\pi)^2} [H(\bar{\alpha}, \mathbf{k}_i^*) - 1] \left[D(m, \mathbf{k}_i^*) k_i^{*\mu} \Big|_{k_0=\omega_{k2,i}^*} \right. \\ \left. + D_{rf}(m, \mathbf{k}_i^*) k_i^{*\mu} \Big|_{k_0=\gamma\sqrt{s_f}+\omega_{P_f k1,i}^*} + D_{ri}(m, \mathbf{k}_i^*) k_i^{*\mu} \Big|_{k_0=\sqrt{s_i}+\omega_{P_i k1,i}^*} \right] \\ - \int \frac{d(\cos \theta_i^*) d|\mathbf{k}_i^*| \mathbf{k}_i^{*2}}{(2\pi)^2} H(\bar{\alpha}, \mathbf{k}_i^*) \left[D_{rf}(m, \mathbf{k}_i^*) k_i^{*\mu} \Big|_{k_0=\gamma\sqrt{s_f}+\omega_{P_f k1,i}^*} + D_{ri}(m, \mathbf{k}_i^*) k_i^{*\mu} \Big|_{k_0=\sqrt{s_i}+\omega_{P_i k1,i}^*} \right]. \quad (13)$$

In what follows we will drop the notation from k_i^* to just k , and similarly with all other quantities, but we advise the reader that everything is evaluated in the frame described by Eq. 10. As an extra simplification we will change the notation in the following sections of the magnitude of the spatial integration momentum from $|\mathbf{k}_i^*|$ to k .

II. REMOVABLE SINGULARITIES

As mentioned before, certain terms of the integrand will be singular within the integration, in this section will look into each of these possible removable singularities from the integrand.

A. Singularities at on-shell conditions $k^{*2} = q^{*2}$

The two-pole propagator $D(m, \mathbf{k})$ can be rewritten in the initial mass frame to be given by

$$D(m, \mathbf{k}) = \frac{1}{2\omega_{k2}(E_f - \omega_{k2} - \omega_{P_f k1})(E_f - \omega_{k2} + \omega_{P_f k1})(E_i - \omega_{k2} - \omega_{k1})(E_i - \omega_{k2} + \omega_{k1})}, \quad (14)$$

$$\omega_{ki} = \sqrt{k^2 + m_i^2}, \quad (15)$$

$$\omega_{P_f k1} = \sqrt{k^2 - 2kP_f \cos \theta + P_f^2 + m_1^2}, \quad (16)$$

where $P_f = |\mathbf{P}_f|$, and we can observe that poles will be present along the integration region whenever the integration variable takes the value of the relative momentum of the initial or final system, i.e. at $k = q_i^*$, and at $k_f^* = q_f^*$. The relative momentum is given by

$$q^*(s, m_1, m_2) = \frac{1}{2\sqrt{s}} \sqrt{s^2 + m_1^4 + m_2^4 - 2(s m_1^2 + s m_2^2 + m_1^2 m_2^2)} = \frac{1}{2\sqrt{s}} \sqrt{(s - (m_1 + m_2)^2)(s - (m_1 - m_2)^2)}. \quad (17)$$

Each of these poles cancels exactly with the term $H(\bar{\alpha}, \mathbf{k}) - 1$, leaving only a finite result.

1. Initial energy on-shell singularity

To find this finite value we take the limit

$$\lim_{k \rightarrow q_i^*} (H(\bar{\alpha}, \mathbf{k}) - 1) D(m, \mathbf{k}) = \frac{-\bar{\alpha}(k_f^{*2} - q_f^{*2})}{4\omega_{k1}\omega_{k2}(E_f - E_i + \omega_{k1} - \omega_{P_f k1})(E_f - E_i + \omega_{k1} + \omega_{P_f k1})} \Big|_{k=q_i^*} \lim_{k \rightarrow q_i^*} \frac{k^2 - q_i^{*2}}{E_i - \omega_{k2} - \omega_{k1}}. \quad (18)$$

where we used that $\lim_{k \rightarrow q_i^*} \omega_{k2} = \lim_{k \rightarrow q_i^*} E_i - \omega_{k1}$. We Taylor expand the denominator around the pole

$$E_i - \omega_{k2} - \omega_{k1} = (k^2 - q_i^{*2}) \frac{-2E_i^3}{E_i^4 - (m_1^2 - m_2^2)^2} + \mathcal{O}((k^2 - q_i^{*2})^2), \quad (19)$$

and we substitute into the previous equation to obtain

$$\lim_{k \rightarrow q_i^*} (H(\bar{\alpha}, \mathbf{k}) - 1) D(m, \mathbf{k}) = \frac{\bar{\alpha}(k_f^{*2} - q_f^{*2})[E_i^4 - (m_1^2 - m_2^2)^2]}{8\omega_{k1}\omega_{k2}E_i^3(E_f - E_i + \omega_{k1} - \omega_{P_f k1})(E_f - E_i + \omega_{k1} + \omega_{P_f k1})} \Big|_{k=q_i^*}, \quad (20)$$

or alternatively

$$\lim_{k \rightarrow q_i^*} (H(\bar{\alpha}, \mathbf{k}) - 1) D(m, \mathbf{k}) = \frac{\bar{\alpha}(k_f^{*2} - q_f^{*2})[E_i^4 - (m_1^2 - m_2^2)^2]}{2E_i^3 2\omega_{k2}(E_f - \omega_{k2} - \omega_{P_f k1})(E_f - \omega_{k2} + \omega_{P_f k1})(E_i - \omega_{k2} + \omega_{k1})} \Big|_{k=q_i^*}, \quad (21)$$

a third option, using the fact that,

$$E_i^4 - (m_1^2 - m_2^2)^2 = 4E_i^2\omega_{k1}\omega_{k2} \Big|_{k=q_i^*}, \quad (22)$$

it somewhat simplifies to

$$\lim_{k \rightarrow q_i^*} (H(\bar{\alpha}, \mathbf{k}) - 1) D(m, \mathbf{k}) = \frac{\bar{\alpha}(k_f^{*2} - q_f^{*2})}{2E_i(E_f - E_i + \omega_{k1} - \omega_{P_f k1})(E_f - E_i + \omega_{k1} + \omega_{P_f k1})} \Big|_{k=q_i^*}. \quad (23)$$

Another option to write minimal amount of code would be

$$\lim_{k \rightarrow q_i^*} \frac{H(\bar{\alpha}, \mathbf{k}) - 1}{2\omega_{k2}[(E_i - \omega_{k2})^2 - \omega_{k1}^2]} = \frac{\bar{\alpha}(k_f^{*2} - q_f^{*2})}{2E_i}. \quad (24)$$

2. Final energy on-shell singularity

Since we evaluate the integral in the initial center of mass frame, the expression for the value of k so that $k_f^* = q_f^*$ is a little more complicated, and similarly the evaluation of the limit. Nonetheless, this condition is equivalent to

$$E_f - \omega_{k2} - \omega_{P_f k1} = 0. \quad (25)$$

The precise definition of k_f^* is given by

$$k_f^* = \left[\sum_{i=1}^3 ([\Lambda_{\beta \mathbf{z}}]_{\mu}^i k^{\mu})^2 \right]^{1/2} \Big|_{k^0 = \omega_{k2}}, \quad (26)$$

where $\beta = (1 - \gamma^{-2})^{1/2}$, the square of which is equal to

$$k_f^{*2} = \gamma^2 k^2 + \beta \gamma^2 (\beta k^2 \cos \theta^2 - 2k \omega_{k2} \cos \theta + \beta m_2^2), \quad (27)$$

in terms of k^2 , $\cos \theta$, γ and m_2 . We will need the derivative with respect to k^2

$$\frac{\partial k_f^{*2}}{\partial k^2} = \gamma^2 + \beta \gamma^2 \left(\beta \cos \theta^2 - \frac{2k^2 + m_2^2}{k \omega_{k2}} \cos \theta \right). \quad (28)$$

To obtain the limit of interest we also need the derivative of the pole associated with the final states on shell. We get a simpler result if we consider the two poles of the $P_f - k$ propagator,

$$\frac{\partial}{\partial k^2} (E_f - \omega_{k2} - \omega_{P_f k1}) (E_f - \omega_{k2} + \omega_{P_f k1}) = -\frac{E_f}{\omega_{k2}} + \frac{P_f \cos \theta}{k}. \quad (29)$$

Combining this equations we obtain

$$\lim_{k_f^* \rightarrow q_f^*} \frac{k_f^{*2} - q_f^{*2}}{(E_f - \omega_{k2})^2 - \omega_{P_f k1}^2} = \frac{\gamma^2 + \beta \gamma^2 \left(\beta \cos \theta^2 - \frac{2k^2 + m_2^2}{k \omega_{k2}} \cos \theta \right)}{-\frac{E_f}{\omega_{k2}} + \frac{P_f \cos \theta}{k}}, \quad (30)$$

$$= \frac{\gamma^2 (-k \omega_{k2} (\beta^2 \cos \theta^2 + 1) + (2\beta k^2 + \beta m_2^2) \cos \theta)}{E_f k - P_f \omega_{k2} \cos \theta}, \quad (31)$$

which further simplifies in the case $\beta = 0$,

$$\lim_{k_f^* \rightarrow q_f^*} \frac{k_f^{*2} - q_f^{*2}}{(E_f - \omega_{k2})^2 - \omega_{P_f k1}^2} \Big|_{\beta=0} = \frac{-\omega_{k2}}{E_f}, \quad (32)$$

which is what we also found for the initial state case.

There is a small caveat in that if $\beta > 0$ and $q_f^* = 0$, both of the derivatives that we calculated vanish for $\cos \theta = 1$, precisely at $k_f^{*2} = 0$. For that case we need the second order derivatives

$$\frac{\partial^2 k_f^{*2}}{\partial (k^2)^2} = \beta \gamma^2 \left(\frac{\omega_{k2}}{2k^3} - \frac{1}{k \omega_{k2}} + \frac{k}{2\omega_{k2}^3} \right), \quad (33)$$

$$\frac{\partial^2}{\partial (k^2)^2} [(E_f - \omega_{k2})^2 - \omega_{P_f k1}^2] = \frac{E_f}{2\omega_{k2}^3} - \frac{P_f}{2k^3}. \quad (34)$$

and we obtain

$$\lim_{k_f^* \rightarrow q_f^*} \frac{k_f^{*2} - q_f^{*2}}{(E_f - \omega_{k2})^2 - \omega_{P_f k1}^2} \Big|_{\cos \theta=1} = \frac{\beta \gamma^2 m_2^4}{E_f k^3 - P_f \omega_{k2}^3}. \quad (35)$$

Then the result for the general case is

$$\lim_{k_f^* \rightarrow q_f^*} \frac{H(\bar{\alpha}, \mathbf{k}) - 1}{(E_f - \omega_{k2})^2 - \omega_{P_f k1}^2} = \frac{\bar{\alpha} (k^2 - q_i^{*2}) \gamma^2 (k \omega_{k2} (\beta^2 \cos \theta^2 + 1) - (2\beta k^2 + \beta m_2^2) \cos \theta)}{E_f k - P_f \omega_{k2} \cos \theta}, \quad (36)$$

for $\beta = 0$ this would reduce to something equivalent to Eq. 24, but replacing $i \rightarrow f$. This case would be important for $q_f^{*2} = 0$ and $\beta = 0$, as otherwise we again obtain a vanishing denominator.

Finally, for $\beta \neq 0$, $\cos \theta = 1$ and $q_f^{*2} = 0$ we get

$$\lim_{k_f^* \rightarrow q_f^*} \frac{H(\bar{\alpha}, \mathbf{k}) - 1}{(E_f - \omega_{k2})^2 - \omega_{P_f k1}^2} \Big|_{\cos \theta=1, q_f^*=0} = \frac{-\bar{\alpha} (k^2 - q_i^{*2}) \beta \gamma^2 m_2^4}{E_f k^3 - P_f \omega_{k2}^3}. \quad (37)$$

3. Simultaneous on-shell conditions

To finalize this section we summarize the combination of the limits when both on-shell conditions happen for the same k ,

$$\lim_{k_f^* \rightarrow q_f^*, k \rightarrow q_i^*} (H(\bar{\alpha}, \mathbf{k}) - 1) D(m, \mathbf{k}) = \frac{\bar{\alpha}}{2E_i} \frac{\gamma^2 (-k \omega_{k2} (\beta^2 \cos \theta^2 + 1) + (2\beta k^2 + \beta m_2^2) \cos \theta)}{E_f k - P_f \omega_{k2} \cos \theta}, \quad (38)$$

which needs to be specialized for the case of $q_f^* = 0$ to avoid evaluating vanishing denominators. For the case of $\beta = 0$ we obtain

$$\lim_{k_f^* \rightarrow q_f^*, k \rightarrow q_i^*} (H(\bar{\alpha}, \mathbf{k}) - 1) D(m, \mathbf{k}) \Big|_{\beta=0, q_f^*=0} = -\frac{\bar{\alpha} m_2}{2E_i E_f}. \quad (39)$$

while for the $\beta \neq 0$, $\cos \theta = 1$ case we obtain,

$$\lim_{k_f^* \rightarrow q_f^*, k \rightarrow q_i^*} (H(\bar{\alpha}, \mathbf{k}) - 1) D(m, \mathbf{k}) \Big|_{\beta \neq 0, q_f^* = 0, \cos \theta = 1} = \frac{\bar{\alpha}}{2E_i} \frac{\beta \gamma^2 m_2^4}{E_f k^3 - P_f \omega_{k2}^3}. \quad (40)$$

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- [1] A. Baroni, R. A. Briceño, M. T. Hansen, and F. G. Ortega-Gama, Form factors of two-hadron states from a covariant finite-volume formalism, Phys. Rev. D **100**, 034511 (2019), arXiv:1812.10504 [hep-lat].